# DR. BENEDIKT KLEINMEIER

**Computer scientist / Passion for good software**
**Interested in hardware and physics / Home-builder**



## CONTACT


kleinmeier.benedikt@gmail.com
Projects
Github
Publications


## SKILLS

*Programming languages*

**Assembler (x86, TriCore)**
**Bash**
**C**
**C++**
**C#**
**Java**
**JavaScript**
**Python**
**Rust**

*Markup languages*

**HTML/CSS**
**LaTeX**

*Operating systems*

**Linux**
**MacOS**
**Windows**

*Software and tools*

**Machine learning**
(Basics of data analysis and LLM development with Jupyter, PyTorch, matplotlib, numpy)
**DevOps**
(Ansible, Docker, Pixi)
**Office**

*Process models*

ISO26262, Scrum, V model

*Development techniques*

Continuous integration, test-driven development

*Languages*

**German**
**English**
**Spanish**

## AWARDS

**2019**
TUM Graduate School Internationalization Grant

**2010**
Nomination scholarship program I.C.S.

**2004**
Best graduate junior high school

## WORK EXPERIENCE

**since 07/2021**
**Infineon Technologies AG, Neubiberg**
Principal software engineer/architect (in-house and customer tools)

- SDK and GUI development (PyQt) of radar applications to configure radar sensors, acquire and visualize data (slides).
- Improving an Eclipse-based application to operate lab equipment and to manage cryptographic keys.
- Development of a Python-based workflow engine including a web dashboard to automatically process customer orders.
- Development of a GUI application (Swing) to control SPICE simulators and to examine simulation results (slides).

C++ | Java | Python | GUI | PyQt | Eclipse/SWT | plotly/dash

**12/2017 – 05/2021**
**Hochschule München / Technical University Munich / University of Sussex, Munich / Brighton**
PhD

- Modeling and simulation of pedestrians and their behavioral changes.
- Implementation and simulation in our own open-source simulator Vadere (`www.vadere.org`).
- GUI programming in Java and data analysis using Python with a focus on test-driven development and continuous integration.

Java | Python | JUnit | pandas | matplotlib | GUI

**05/2015 – 11/2017**
**Infineon Technologies AG, Neubiberg**
Expert firmware developer

- Development of bootstrap loaders and startup code for various microcontrollers in C.
- Establishing systematic unit testing to meet ISO 26262 requirements.

C | Python | Ceedling

## STUDIES

**10/2012 – 01/2015**
**Hochschule München, Munich**
Computer science (M.Sc) — embedded systems focus

- Master thesis at Fraunhofer ESK: Development of a software solution for highly accurate time and position detection for Car2X scenarios
- Focus: Driver development under GNU/Linux in C

C | C++

**10/2008 – 09/2012**
**Hochschule München, Munich**
Computer science (B.Sc)

- Bachelor thesis at Infineon Technologies AG: Development of an evaluation tool for barometric air pressure sensors
- Focus: Firmware development for 8- and 16-bit microcontrollers in C and PC software in C#/.NET

C | C# | .NET

## ACTIVITIES BEFORE STUDIES

**03/2008 – 09/2008**
**SEP AG (backup software), Weyarn**
Support staff

- Setting up test environments (Windows, GNU/Linux), software testing, maintaining user documentation.
  GNU/Linux | Windows | Documentation

**03/2007 – 11/2007**
**Caritas St.-Anna-Haus, Holzkirchen**
Community service worker in retirement home

- Nursing and housekeeping

## SCHOOL

**2004 – 2006**
**Technical secondary school and vocational secondary school, Bad Tölz**
Focus: Economy and management
Two internships: local bank (departments: human resources, accounting, credit) and district office Miesbach

**2000 – 2004**
**Junior high school, Miesbach**

**1994 – 2000**
**Elementary school, Weyarn**

## PUBLICATIONS

Peer-Reviewed journal articles:

- Benedikt Kleinmeier, Gerta Köster, and John Drury. "Agent-Based Simulation of Collective Cooperation: From Experiment to Model". In: *Journal of the Royal Society Interface* 17 (171 2020), p. 20200396. ISSN: 1742-5662. DOI: 10.1098/rsif.2020.0396. URL: https://arxiv.org/abs/2005.12712
- Benedikt Kleinmeier et al. "Vadere: An Open-Source Simulation Framework to Promote Interdisciplinary Understanding". In: *Collective Dynamics* 4 (2019). DOI: 10.17815/CD.2019.21

October 30, 2026
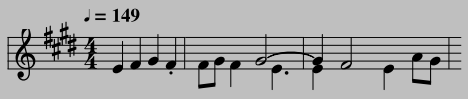


**Benedikt Kleinmeier**

## HOBBIES

Freestyle football | Football | GNU/Linux | Mnemotechnics | Playing the piano | Music | Mountainbiking | Hiking

## MY PHILOSOPHY

- Joy of learning: That's why I have been learning to play the piano autodidactically.




- Work-life balance and nature: For this reason, I have crossed the alps with my mountain bike.

